Introduction to Beamer

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Presentation Outline

1. Introduction
2. Core Features
3. Technical Content
4. Workflow

What is Beamer?

- Beamer is a LaTeX document class for creating slides.
- Slides are called **frames**.
- A source file is compiled to a portable PDF.
- Ideal for mathematical and technical presentations.

Frames and Structured Lists

1. Declare a frame with `begin{frame}`.
2. Add a title and content.
3. Close with `end{frame}`.

Blocks and Emphasis

Standard Block

Group related ideas in a named block.

Important

Keep slides concise and readable.

Example

Use `\textbf` for bold text.

Overlays: Step-by-Step Reveal

- First idea appears on overlay 1.

Overlays: Step-by-Step Reveal

- First idea appears on overlay 1.
- Second idea appears on overlay 2.

Overlays: Step-by-Step Reveal

- First idea appears on overlay 1.
- Second idea appears on overlay 2.
- Third idea appears on overlay 3.

Mathematical Typesetting

The quadratic formula is

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \qquad a \neq 0.$$

An inline expression: $E = mc^2$.

Tables in Beamer

Feature	Purpose
Frames	Organize slides
Themes	Control presentation appearance
Overlays	Reveal content progressively
Blocks	Highlight related information

Table: Common Beamer features

Two-Column Layout

Advantages

- Consistent design
- Excellent equations
- Version-control friendly

Considerations

- Requires LaTeX syntax
- Compilation needed
- Fine-grained visual design can take time

Practical Workflow

1. Write a `.tex` source file.
2. Select a theme and add frames.
3. Compile using `pdflatex` or Overleaf.
4. Review the PDF and fix any errors.
5. Commit both source and final output to GitHub.

Learning Outcomes

- Create slides using the Beamer class.
- Use themes, blocks, columns, equations, and tables.
- Understand overlay specifications.
- Compile a reproducible PDF presentation.

Thank You!

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